# HF16

16 Mbit (2 MB) serial NOR flash memory

## Features

- Supply voltage: 2.7 V to 3.6 V
- Clock frequency up to 104 MHz (Fast Read); Dual and Quad I/O supported
- 100,000 program/erase cycles
- Data retention: 20 years
- Page program time: 0.7 ms typical (3 ms maximum)
- 4 KB sector erase: 45 ms typical (400 ms maximum)
- Chip erase: 8 s typical (30 s maximum)
- Standby current: 15 µA; deep power-down: 1 µA
- WSON-8 (6×5 mm) and SOIC-8 packages
- Industrial grade –40 °C to 85 °C; automotive HF16-A –40 °C to 105 °C

## General description

The HF16 is a 16 Mbit SPI NOR flash with a unique 64-bit ID, security registers and software write protection.

## Applications

Firmware storage, Boot code, Data logging, Configuration storage.

## Ordering information

| Order code | Package | Packing | Temperature grade |
|---|---|---|---|
| HF16-WSON8-T | WSON-8 (6×5 mm) | Tape & reel, 3000 | –40 °C to 85 °C |
| HF16-SOIC8-T | SOIC-8 | Tape & reel, 2500 | –40 °C to 85 °C |
| HF16A-WSON8-T | WSON-8 (6×5 mm) | Tape & reel, 3000 | –40 °C to 105 °C |

## Absolute maximum ratings

Stresses beyond these ratings may cause permanent damage. Functional operation is not implied.

| Parameter | Min | Max | Unit |
|---|---|---|---|
| VCC to GND | –0.5 | 4.2 | V |
| Voltage on any pin | –0.5 | VCC + 0.4 | V |
| Storage temperature | –65 | 150 | °C |

## Recommended operating conditions

| Parameter | Min | Typ | Max | Unit |
|---|---|---|---|---|
| Supply voltage VCC | 2.7 | 3.3 | 3.6 | V |
| SPI clock (Fast Read) | — | — | 104 | MHz |
| Operating temperature (industrial) | –40 | — | 85 | °C |

## Electrical characteristics

Typical values at TA = 25 °C unless otherwise noted.

| Parameter | Conditions | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| Endurance | Program/erase cycles | 100000 | — | — | cycles |
| Data retention | At 55 °C | 20 | — | — | years |
| Page program time | 256 bytes | — | 0.7 | 3 | ms |
| Sector erase time | 4 KB | — | 45 | 400 | ms |
| Chip erase time | | — | 8 | 30 | s |
| Standby current | CS high | — | 15 | 30 | µA |
| Deep power-down current | | — | 1 | 5 | µA |

## Pin configuration and functions

| Pin | Name | Type | Description |
|---|---|---|---|
| 1 | CS# | Input | Chip select, active low. |
| 2 | DO/IO1 | I/O | Data out / IO1. |
| 3 | WP#/IO2 | I/O | Write protect / IO2. |
| 4 | GND | Ground | Ground. |
| 5 | DI/IO0 | I/O | Data in / IO0. |
| 6 | CLK | Input | Serial clock. |
| 7 | HOLD#/IO3 | I/O | Hold / IO3. |
| 8 | VCC | Power | 2.7 V to 3.6 V supply. |

## Package and thermal information

| Package | Thermal resistance θJA | Dimensions |
|---|---|---|
| WSON-8 (6×5 mm) | θJA = 45 °C/W | 6.0 × 5.0 mm |
| SOIC-8 | θJA = 125 °C/W | 5.0 × 4.0 mm |

## Application information

Use the HM3300 regulator to provide a clean 3.3 V supply to the flash and microcontroller. Pull CS# up to VCC during power-up.

## Revision history

| Revision | Date | Change |
|---|---|---|
| Rev A | 2023-08-30 | Initial release |
| Rev B | 2024-06-12 | Added automotive grade HF16-A |